Econometrics I

Lecture 9a: Difference-in-Differences

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Fall 2026

Where we are

- Last week: the panel event study is a stack of 2×2 comparisons, one per event time, sharing the period before the event as a baseline.
- Today, Lecture 9a: the 2×2 itself.
  - What it identifies, and the one assumption that buys it.
  - When to believe that assumption, and what breaks it.
  - How to do inference when treatment is assigned to groups.
- Lecture 9b: what goes wrong when different units are treated at different dates, and the fixes.
- Running example: a disclosure mandate that hits large firms and exempts small ones. Simulated, so we always know the right answer.

How do we construct the counterfactual?

We observe $Y_{it}(D_{it})$ but never $Y_{it}(1 - D_{it})$. Four ways to fill in the missing cell:

Before–after: use the treated unit's own past, $Y_{i,t-1}(0)$. Fails if anything else changed over time.

Matching: use a different unit j with $X_j = X_i$ at the same date. Fails if they differ in unobservables.

Difference-in-differences: use the *change* in other units, $Y_{jt}(0) - Y_{j,t-1}(0)$, as the counterfactual change for i. Levels can differ; trends cannot.

Synthetic control: use a weighted average of other units chosen to track i before treatment (Session 11).

The two-by-two

Setup

- Two groups, $D_i \in \{0, 1\}$; two periods, $t \in \{1, 2\}$. The treated group is treated in period 2 only: $D_{it} = D_i \cdot Post_t$.

	Treated group ($D_i = 1$)	Control group ($D_i = 0$)
Period 1 (before)	$Y_{i1}(0)$	$Y_{i1}(0)$
Period 2 (after)	$Y_{i2}(1)$	$Y_{i2}(0)$

- The target is the effect on the treated in period 2,

$$\text{ATT} = \mathbb{E}\left[\, Y_{i2}(1) - Y_{i2}(0) \mid D_i = 1 \,\right].$$

- The missing cell is $\mathbb{E}[Y_{i2}(0) \mid D_i = 1]$: what the treated group would have done without treatment.

Two differences

Write the change in the outcome as $\Delta Y_i = Y_{i2} - Y_{i1}$. For the treated group:

$$\mathbb{E}[\Delta Y_i \mid D_i = 1] = \overbrace{\mathbb{E}\left[Y_{i2}(1) - Y_{i2}(0) \mid D_i = 1\right]}^{\text{ATT}} + \overbrace{\mathbb{E}\left[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 1\right]}^{\text{trend}(1)}$$

For the control group:

$$\mathbb{E}[\Delta Y_i \mid D_i = 0] = \overbrace{\mathbb{E}\left[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 0\right]}^{\text{trend}(0)}$$

The difference in differences:

$$\Delta_{DD} = \mathbb{E}[\Delta Y_i \mid D_i = 1] - \mathbb{E}[\Delta Y_i \mid D_i = 0] = \text{ATT} + \big[\text{trend}(1) - \text{trend}(0)\big].$$

Parallel trends

- Δ_{DD} = ATT exactly when the bracket is zero:

$$\mathbb{E}\left[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 1\right] = \mathbb{E}\left[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 0\right].$$

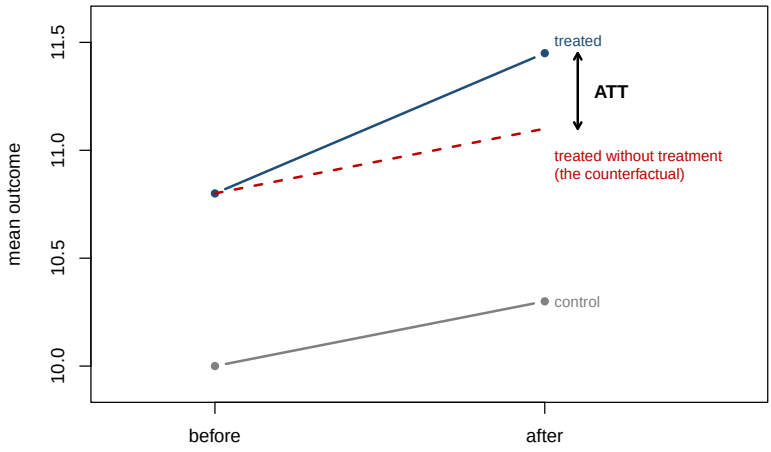
  Absent treatment, both groups would have changed by the same amount on average.

- What it allows:
  - Different levels: $\mathbb{E}[Y_{it}(0) \mid D_i = 1] \neq \mathbb{E}[Y_{it}(0) \mid D_i = 0]$. Selection on anything permanent is fine.
  - Selection on gains: the treated may benefit more than the controls would. That is why the target is the ATT, not the ATE.

- What it rules out: anything that moves the two groups' untreated outcomes *differently* between the periods.

- It is an assumption about a counterfactual. No data on the post period can test it.

The picture


mean outcome
11.5
11.0
10.5
10.0
before
after
treated
ATT
treated without treatment
(the counterfactual)
control

Running example: an internal-control mandate

Running example: SOX Section 404

- The Sarbanes–Oxley Act (July 2002), Section 404: management must report on internal controls and the auditor must attest to that report.
- Accelerated filers (public float $\geq$ \$75M) first complied for fiscal years ending on or after November 15, 2004. Smaller firms were exempt for years.
- Question: what did compliance do to audit fees? (For real firms near the threshold, Iliev 2010 estimates the management report raised audit fees by 98%.)
- Our simulation: 2,000 firms, fiscal 2000–2007, log audit fees.
  - Firm effects (big firms pay more) and common year shocks.
  - True effect on accelerated filers: $+0.35$ in 2004, decaying to $+0.20$ by 2007 as auditors learn.
  - One complication, on purpose: larger firms are on a steeper audit-fee trend.

Four ways to compute the same number

Take fiscal 2003 (before) and 2004 (after). $D_i = 1$ for accelerated filers.

1. Cell means: $(\bar{y}_{T,04} - \bar{y}_{T,03}) - (\bar{y}_{C,04} - \bar{y}_{C,03})$.
2. First differences: regress $\Delta y_i = y_{i,04} - y_{i,03}$ on a constant and D_i: $\Delta y_i = \beta_{post} + \delta \cdot D_i + \Delta\varepsilon_i$.
3. Interaction regression on the stacked panel: $y_{it} = \beta_0 + \alpha \cdot D_i + \gamma \cdot Post_t + \delta \cdot D_i \times Post_t + \varepsilon_{it}$.
4. Two-way fixed effects: $y_{it} = \alpha_i + \gamma_t + \delta \cdot D_{it} + \varepsilon_{it}$.

In (3), α is the level gap in 2003, γ the control group's change, δ the DiD.

Route	Estimate	Std. error
(1) Cell means: $(\bar{y}_{T,04} - \bar{y}_{T,03}) - (\bar{y}_{C,04} - \bar{y}_{C,03})$	0.4363	—
(2) First differences: Δy_i on D_i (robust)	0.4363	(0.008)
(3) Interaction: y_{it} on D_i, $Post_t$, $D_i \times Post_t$ (cluster firm)	0.4363	(0.008)
(4) Two-way FE: y_{it} on D_{it} | firm, year (cluster firm)	0.4363	(0.008)
True effect on accelerated filers in 2004	0.3500	

Why four routes?

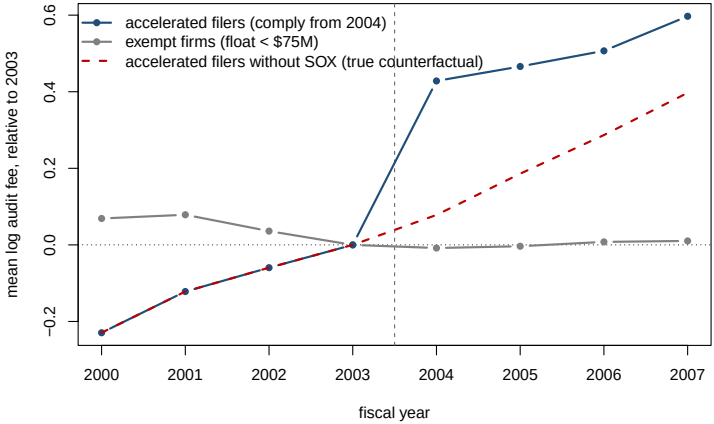
- With two periods and a balanced panel they are numerically identical: same point estimate, and the same standard error once you cluster by firm in (3) and (4).
- Each one is useful for something:
  - (1) is what you put in a table and explain to a manager.
  - (2) makes the identifying assumption visible: $\mathbb{E}[\Delta\varepsilon_i \mid D_i] = 0$ *is* parallel trends.
  - (3) gives you the level gap α, which is *allowed* to be large.
  - (4) extends to many periods, many units and covariates. It is also the version that breaks with staggered timing (Lecture 9b).
- But our estimate is 0.436 and the truth is 0.35. The arithmetic was fine. The assumption was not.

Every comparison, against the truth

Comparison	Estimate	Std. error	Truth
Post-period gap: accelerated — exempt, 2004	1.245	(0.024)	0.35
Before–after, accelerated filers, 2003→2004	0.428	(0.006)	0.35
Difference-in-differences	0.436	(0.008)	0.35
DiD with size × year controls	0.328	(0.013)	0.35

- Post-period gap confounds the effect with the permanent level gap: big firms always paid more.
- Before–after confounds it with whatever happened to all audit fees that year.
- DiD removes both, but not a *difference* in trends. Accelerated filers are big, and big firms' fees were already growing faster.

What went wrong: big firms were already pulling away



The red dashed line is the treated group's true counterfactual. The exempt group is not tracking it.

Conditional parallel trends

Conditional parallel trends

- Parallel trends may fail unconditionally but hold *within* groups of similar firms:

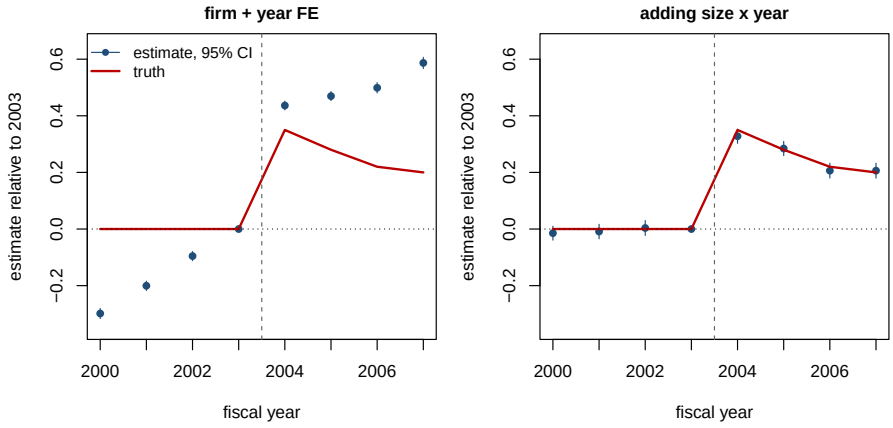
$$\mathbb{E}\left[Y_{i2}(0) - Y_{i1}(0) \mid X_i, D_i = 1\right] = \mathbb{E}\left[Y_{i2}(0) - Y_{i1}(0) \mid X_i, D_i = 0\right].$$

- It is the matching assumption applied to the *change* in the outcome.
- Simplest implementation: let the year effects depend on pre-treatment covariates,

$$y_{it} = \alpha_i + \gamma_t + \textstyle\sum_s \beta_s \cdot \mathbb{1}\{t = s\} \cdot X_i + \delta \cdot D_{it} + \varepsilon_{it}.$$

- With two periods this is the first-difference regression with X_i added: Δy_i on D_i and X_i.
- In the SOX example, X_i = firm size. Size $\times$ year moves the estimate from 0.436 to 0.328 (truth 0.35).

The event study shows the problem and the fix



Left: the pre-period coefficients trend up into 2003; the post-period ones inherit the trend. Right: size-specific year effects flatten the pre-period and put the post-period on the truth.

In practice

```
fit_2x2  <- feols(y ~ D | firm + year, sox[year %in% 2003:2004], vcov = ~firm)      # the 2x2
fit_es   <- feols(y ~ i(year, acc, ref = 2003) | firm + year, sox, vcov = ~firm)    # event study
fit_cond <- feols(y ~ i(year, acc, ref = 2003) + i(year, size, ref = 2003) | firm + year,
                  sox, vcov = ~firm)                                                # + size x year
iplot(list(fit_es, fit_cond))
```

```
fit_cond = pf.feols("y ~ i(year, acc, ref=2003) + i(year, size, ref=2003) | firm + year",
                    data=sox, vcov={"CRV1": "firm"})
fit_cond.iplot()
```

- $\mathrm{i}(\mathrm{year},\ \mathrm{size})$ interacts each year dummy with a *continuous* variable: one size slope per year.
- Use covariates measured *before* treatment. Here size is fixed at its pre-SOX value.

Conditioning needs overlap

- Why not size-decile $\times$ year fixed effects? Because treatment is assigned *by* size. Every decile except the one containing $75M is all treated or all exempt.
- Within-cell comparisons then exist only for firms near the cutoff. That is a regression discontinuity (Session 10), and it is what Iliev (2010) estimates.
- The linear size $\times$ year model works here because the simulated size trend really is linear. It extrapolates the small firms' size–trend relationship to the big ones.
- General lesson: conditional parallel trends needs treated and control units at the same values of X. When it fails, you are relying on the functional form.

Covariates done right

- Time-invariant, pre-treatment X_i interacted with time: what we just did. Safe.
- Time-varying X_{it} that treatment can move is a bad control. SOX changes internal-control spending, audit hours and auditor choice. Controlling for any of them blocks part of the effect (mediator) or opens a collider path (Session 5).
- Rule of thumb: condition on covariates measured *before* treatment, or on ones the treatment cannot plausibly affect.
- Doubly robust DiD (Sant'Anna and Zhao 2020): combine an outcome regression for ΔY among controls with a propensity score for D_i given X_i. It is consistent if either one is right. That is AIPW (Session 11) applied to the change in the outcome. In R: DRDID, or $\mathrm{did::att_gt}(\ldots, \mathrm{est_method} = \text{"dr"})$.

The 2×2 that made DiD famous: Card and Krueger (1994)

TABLE 3—AVERAGE EMPLOYMENT PER STORE BEFORE AND AFTER THE RISE IN NEW JERSEY MINIMUM WAGE

	Stores by state			Stores in New Jersey[a]			Differences within NJ[b]	
Variable	PA (i)	NJ (ii)	Difference, NJ − PA (iii)	Wage = $4.25 (iv)	Wage = $4.26–$4.99 (v)	Wage ≥ $5.00 (vi)	Low-high (vii)	Midrange-high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	−2.69 (1.37)	−2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)	0.75 (1.27)
3. Change in mean FTE employment	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)	1.32 (0.95)	0.87 (0.84)	−2.04 (1.14)	3.36 (1.48)	2.91 (1.41)
4. Change in mean FTE employment, balanced sample of stores[c]	−2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.21 (0.82)	0.71 (0.69)	−2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0[d]	−2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	−2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

Notes: Standard errors are shown in parentheses. The sample consists of all stores with available data on employment. FTE (full-time-equivalent) employment counts each part-time worker as half a full-time worker. Employment at six closed stores is set to zero. Employment at four temporarily closed stores is treated as missing.

[a]Stores in New Jersey were classified by whether starting wage in wave 1 equals $4.25 per hour ($N = 101$), is between $4.26 and $4.99 per hour ($N = 140$), or is $5.00 per hour or higher ($N = 73$).

[b]Difference in employment between low-wage ($4.25 per hour) and high-wage ($\geq$ $5.00 per hour) stores; and difference in employment between midrange ($4.26–$4.99 per hour) and high-wage stores.

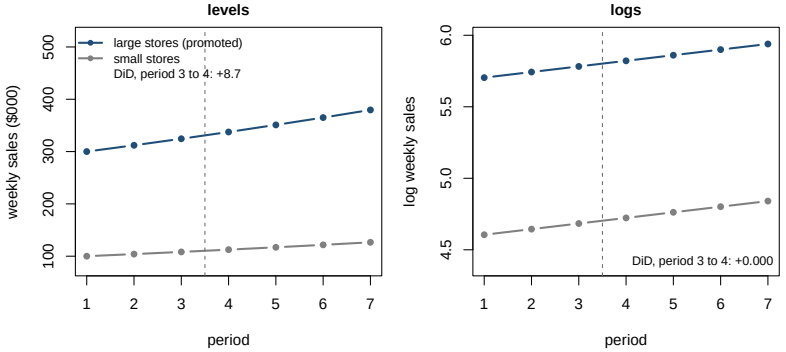
[c]Subset of stores with available employment data in wave 1 and wave 2.

[d]In this row only, wave-2 employment at four temporarily closed stores is set to 0. Employment changes are based on the subset of stores with available employment data in wave 1 and wave 2.

- April 1992: New Jersey raises its minimum wage from $4.25 to $5.05. Pennsylvania does not.
- Fast-food restaurants surveyed before and after. Compare the change in employment.
- Employment did not fall in NJ relative to PA, contrary to the textbook prediction.
- A long debate followed over the data source (Neumark and Wascher 2000; Card and Krueger 2000).
- One treated state and one control state. Hold that thought for inference.

What parallel trends is and isn't

Parallel trends depends on the scale



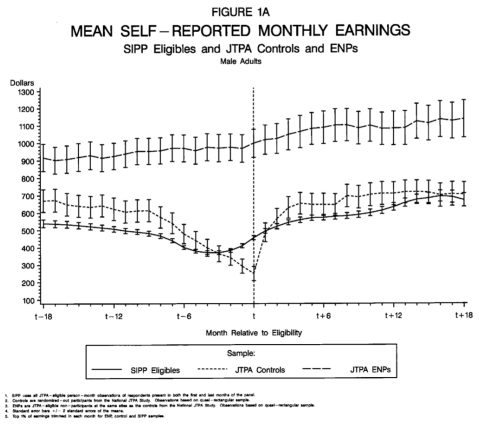
- Both store types grow 4% a period and the promotion does nothing. Trends are parallel in logs, so they cannot be parallel in levels.
- It can hold in every monotone transformation only if it holds for the whole distribution of $Y(0)$ (Roth and Sant'Anna 2023). Decide on substance whether the counterfactual change is additive or proportional.

Flat pre-trends are not parallel trends

- From Session 8: a pre-period that looks flat is reassuring, not proof.
  - Pre-tests have low power against the violations that matter (Roth 2022).
  - Report what the pre-period rules out, and how the estimate moves if post-period violations are as large as pre-period ones (Rambachan and Roth 2023; HonestDiD).
- Compound treatments: parallel trends is about what would have happened *at the date of treatment*. Anything else that changed in 2004 for large firms but not small ones is in the estimate. Examples: other rules phased in by size, exchange listing standards, a change in how large audit firms priced.
- The question to ask: *why* did the treated group get treated *then*? If the answer involves the outcome, worry.

Threats to the design

The Ashenfelter dip (Heckman and Smith 1999)



Training participants' earnings fall just before they enrol. They enrol *because* of a bad draw.

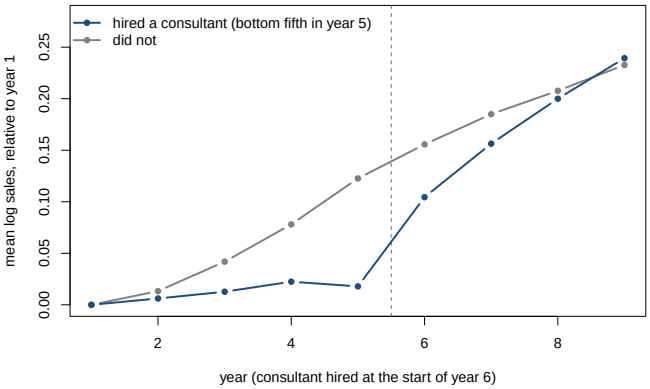
Ashenfelter dip: algebra

- Model $Y_{it}(0) = \alpha_i + \gamma_t + u_{it}$, with u_{it} i.i.d. over time. Units enter treatment in period 2 if $Y_{i1} < B$, so $u_{i1} < B - \alpha_i - \gamma_1$.
- The permanent part α_i differences out. The *transitory* part does not:

$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 1] = \gamma_2 - \gamma_1 - \mathbb{E}[u_{i1} \mid u_{i1} < B - \alpha_i - \gamma_1],$$
$$\mathbb{E}[Y_{i2}(0) - Y_{i1}(0) \mid D_i = 0] = \gamma_2 - \gamma_1 - \mathbb{E}[u_{i1} \mid u_{i1} \geq B - \alpha_i - \gamma_1].$$

- The difference is $\mathbb{E}[u_{i1} \mid u_{i1} \geq c_i] - \mathbb{E}[u_{i1} \mid u_{i1} < c_i] > 0$, writing $c_i = B - \alpha_i - \gamma_1$. DiD overstates the effect.
- Regression to the mean: the unlucky recover whether or not they are treated.
- Business versions: firms hire a consultant, replace a CEO, or adopt a new system *after* a bad quarter.

Simulated: a turnaround consultant who does nothing



Firms in the bottom fifth of year-5 sales hire a consultant. Its true effect is zero. Shocks are AR(1) with $\rho = 0.6$.

Base year	DiD, year 7 vs base	Truth
Year 5 (the bad year)	0.076	0
Year 3	0.001	0
Year 1	-0.029	0

The bias depends on the base year and can change sign. Which base year is right depends on shock dynamics you do not observe. Model the selection rule instead.

More threats, in SOX terms

- Composition. Treatment changes who is in each group.
  - Some firms went private or deregistered to escape SOX costs (Engel, Hayes and Wang 2007; Leuz, Triantis and Wang 2008).
  - Some kept their public float below $75M to stay exempt (Gao, Wu and Zimmerman 2009).
  - Fix: define groups by pre-period characteristics (float in 2002), keep a balanced panel, and report attrition by group.
- Anticipation. The Act passed in 2002 and the rules were set in 2003–04. Firms that start paying for compliance in 2003 contaminate the base year. Use an earlier base period.
- Spillovers onto controls. If auditors are capacity-constrained, compliance work for big clients raises fees for small clients too. The control group is then partly treated, and DiD understates the effect (the interference problem from Session 5).
- Selection on the outcome. The Ashenfelter dip, again: here the timing is set by law, which is exactly why it is a good design.

Inference

Serial correlation: placebo laws (Bertrand, Duflo and Mullainathan 2004)

- A state-by-year panel of log e-commerce sales, 20 years, AR(1) errors. Half the states "pass a law" in a random year. The law does nothing. Rejection rates of a nominal 5% test, 1,000 simulations:

Standard errors	50 states $\rho = 0$	50 states $\rho = 0.8$	10 states $\rho = 0.8$
Conventional (iid)	6%	38%	37%
Heteroskedasticity-robust	6%	38%	37%
Clustered by state	7%	6%	6%

- With persistent errors, the treatment dummy is persistent too. Conventional SEs treat 20 correlated years as 20 independent draws.
- Cluster at the level at which treatment is assigned (the state), so each state's whole time series counts as one draw. That needs enough clusters, and enough *treated* clusters.

Inference with few treated clusters

- Card and Krueger's SEs treat 410 stores as independent, but the policy varies by *state*: NJ vs. PA.
- Stores in a state share shocks (local demand, weather). Cluster-robust SEs need many clusters, and many *treated* clusters. With $G = 2$ you cannot cluster by state at all.
- The wild cluster bootstrap (Cameron, Gelbach and Miller 2008) fixes *few clusters*, not *few treated clusters*. With one or two treated clusters it still over-rejects (MacKinnon and Webb 2018).
- With one treated group and *many* control groups:
  - Conley and Taber (2011): the control groups' residuals estimate the distribution of the treated group's error.
  - Randomization inference: assign the policy to each control group in turn (the placebo logic of Session 11).
  - Aggregate to group $\times$ period and use $G - 1$ degrees of freedom (Donald and Lang 2007).
- With one control group, as in Card and Krueger, none of these can deliver policy-level inference. Count your treated (and control) clusters before you report a standard error.

Variants and applications

Triple differences

- Example: a state privacy law restricts ad targeting. Compare in-state vs. out-of-state (S), before vs. after (P), and ad-supported vs. subscription apps (G). Subscription apps should be unaffected.

$$y = \beta_0 + \beta_1 S + \beta_2 G + \beta_3 P + \beta_4 SG + \beta_5 SP + \beta_6 GP + \delta\, SGP + \varepsilon$$

- $\hat{\delta}$ is the DiD for ad-supported apps minus the DiD for subscription apps. Include *every* two-way interaction (or the corresponding fixed effects).
- Identification needs only one parallel-trends assumption: the *gap* between the two app types' trends would have been the same in and out of state. Neither DiD needs to be unbiased on its own (Olden and Møen 2022).
- It is weaker than parallel trends, not free: a shock that hits only ad-supported apps in the law state still biases it.

DiD in your fields

- Accounting.
  - Mandatory IFRS adoption around the world and its liquidity and cost-of-capital effects (Daske, Hail, Leuz and Verdi 2008, *JAR*).
  - SOX 404 costs and earnings quality at the $75M threshold (Iliev 2010, *JF*).
  - Firms managing their float to stay exempt (Gao, Wu and Zimmerman 2009, *JAR*): the composition threat in action.
- Marketing / IS.
  - Book reviews on two retailers' sites: the change in a title's relative sales rank against the change in its relative reviews (Chevalier and Mayzlin 2006, *JMR*).
  - A city soda tax: prices, cross-border shopping and purchases relative to untaxed comparison areas (Seiler, Tuchman and Yao 2021, *JMR*).
- Reading one: find the 2×2, name the control group, and ask what else changed for the treated group at that date.

Toward Lecture 9b: SOX was phased in

- Accelerated filers: full Section 404 from fiscal years ending on or after November 15, 2004.
- Smaller firms: the management report from fiscal years ending on or after December 15, 2007.
- The auditor attestation for small firms never arrived. Dodd–Frank (2010) exempted them.
- That is staggered adoption: different cohorts treated at different dates. The natural regression is the same two-way FE model with D_{it} switching on at each cohort's date.
- It now makes comparisons we never asked for, including *already-treated* firms serving as controls. When effects evolve over time, as they do here (0.35 decaying to 0.20), those comparisons are biased.
- Lecture 9b: what that regression estimates, and the estimators that fix it.

Recap

- DiD replaces “same level” with “same *change*”. Permanent differences are fine; differential changes are not.
- In a two-period panel, the four routes (means, first differences, interaction, two-way FE) give the same number. That number is right only if parallel trends holds.
- Parallel trends is scale-dependent and untestable. Pre-periods are diagnostics. Conditioning on pre-treatment X helps when there is overlap.
- Threats: selection on transitory shocks, composition, anticipation, spillovers, compound treatments.
- Inference: cluster at the level of assignment, and count the treated clusters.

Workbook time

Fall/L9 - Quasi Experiments/inclass_session9.Rmd, Part 1 (Python: inclass_session9_py.qmd):

1. Reproduce the four routes on the SOX simulation. Then turn off the size trend: what happens to the gap between DiD and the truth?
2. Add size $\times$ year controls. Replace them with size-decile $\times$ year fixed effects. What happens to the standard error, and why?
3. Run the placebo-law simulation. Vary ρ and the number of states, and find where clustering stops working.